

NETWORKING 01

What is Networking?

👉 **Networking = devices talking to each other**

Just like:

- You send a message → your friend receives it
- Computer sends data → another computer receives it

✓ Devices can be:

- Laptop
- Mobile
- Server
- Router

How do they talk?

They use 3 main things:

1. IP Address (Identity)

👉 Like your home address

- Every device has a unique number
- Example: 192.168.1.1

📌 Without IP → devices can't find each other

2. Protocols (Rules)

👉 Like language or grammar

- Rules for communication
- Example:
 - HTTP → websites
 - HTTPS → secure websites
 - FTP → file transfer

📌 Without rules → communication becomes chaos



1. HTTP (HyperText Transfer Protocol)

👉 Used to open websites

Simple meaning:

- It is the rule your browser uses to talk to a website server

Example:

```
http://example.com
```

What it does:

1. You request a webpage
2. Server sends back data (HTML, images)

📌 Problem:

- ❌ Not secure
- Anyone can see the data (hackers on same network)



2. HTTPS (Secure version of HTTP)

👉 Same as HTTP but **SAFE**

Example:

```
https://google.com
```

What is extra?

- 🔒 Encryption (data is locked)
- No one can read your data

Real-life:

- Passwords
- Banking
- Payments

📌 Always prefer HTTPS

3. FTP (File Transfer Protocol)

👉 Used to transfer files between computers

Example uses:

- Uploading website files to a server
- Downloading large files

How it works:

- One computer = client
- Other = server
- Files move between them

📌 Problem:

- ❌ Not secure (normal FTP)

3. Packets (Small pieces of data)

👉 Like breaking a big parcel into small boxes

- Data is NOT sent as one big file
- It is split into **small packets**
- Sent → then reassembled at destination

📌 Why? Faster + reliable

Is packet-based communication same in undersea optical fiber cables?

👉 YES ✅ — exactly the same concept is used

But only one thing changes:

👉 **the medium (how data travels)**

✓ Packet = Data + Address info (source & destination)



Core Idea (VERY IMPORTANT)

No matter where data goes:

- WiFi 📶
- Mobile network 📱
- Optical fiber 🌊

👉 **Data is ALWAYS sent as packets**